



Introduction to Generative AI

CSIT205 Generative AI – 2025



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Wollongong
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*SCHOOL OF COMPUTING AND INFORMATION
TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION
SCIENCES*



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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi
Artist Lauren Henry.

The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, partially cut off. The 'U' is at the top, the 'O' is in the middle, and the 'W' is at the bottom.

Everyone is welcome here.

UOW takes pride in
the diversity of our
community.

Find out more, seek support or join
the UOW Ally Network
uow.info/ally

Stay in the know...

Keep connected at UOW

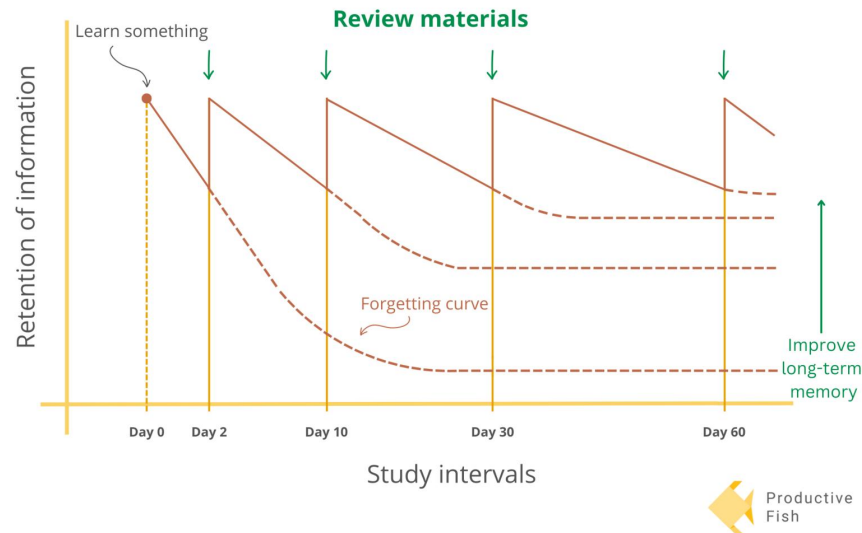
Discover the student systems and communication channels you'll use both on campus and at home.

UOWMail
SOLS
AskUOW
MyUOW
Moodle
Instagram
Website



We will have breaks and repetitions

- Why? Your brain is like a muscle 🦵 it grows (read: learns) at rest
 - This is why cramming does not work
- How? Learn - space - repeat - space - repeat
 - Vary in space size
 - Longer spaces generally work better; 1 hour is enough
 - Make sure to do something completely different during the 'spaces'



Subject introduction



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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Topic	Labs	Readings	Task due
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Module 1: Generative AI and Assisted Coding

Introduction to Generative AI			
Text and Image Generation with Generative AI		Chapter 3 & 4	
Understanding Assisted Coding	Lab 1	Poldrack Lu, & Begus, 2023	
Advanced Topics and Emerging Trends in Generative AI and Assisted Coding		Chapter 10 & 13	Quiz

Module 2: Enhancing Model Performance

Introduction to Prompt Engineering		Wang et al., 2023; Wei et al., 2022	Assessment 1
Retrieval Augmented Generation (RAG)			
Fine-Tuning and Reinforcement Learning with Human Feedback (RLHF):	Lab 2	Chapter 9	
Advanced Topics and Emerging Trends in Enhancing Model Performance			

Module 3: Business Applications and Strategy

Business Vision Setting for Generative AI			Assessment 2
Strategy for Implementing Generative AI in Business	Lab 3		
Data Management, Privacy, and Ethics		Weidinger et al., 2021	
Advanced Topics and Emerging Trends in Business Applications and Strategy	Lab 4		
Wrap-up and summary			

Assessment overview

No.	Assessment Name	Assessment Weight	Mapping to Subject Learning Outcome	Task Due
1	Individual Quiz	0%	1	Session Week 4
2	Assessment 1 (Prompt engineering)	25%	2	Session Week 6
3	Assessment 2 (Business case)	25%	4	Session Week 8
4	Final exam	50%	1, 2, 3, 4, 5	

Assessment 1

- Deliverables (For both reports, write your full name and UOW ID at the top of first page)
- File name must be in the form of: TXX_NAME_UOWID.txt for prompt-answer pairs where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (not SIM student ID number).

For example, T02_BobChew_8080426.txt

Assessment 1

- Deliverables (For both reports, write your full name and UOW ID at the top of first page)
- File name must be in the form of: TXX_NAME_UOWID.doc for report where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (not SIM student ID number).
For example, T02_BobChew_8080426.doc

Assessment 2

Assessment 2 – Generative AI business case

- (Day 1 – Form a group of 5 or 6 students and select a responsible student as a group leader. The group leader of a tutorial class is to inform the lecturer of the tutorial class, group members UOW id and names before end of 8th day.)
- (All team members must be present unless there is a valid reason during the interview with CTO during lab 3 and presentation during lab 4.)

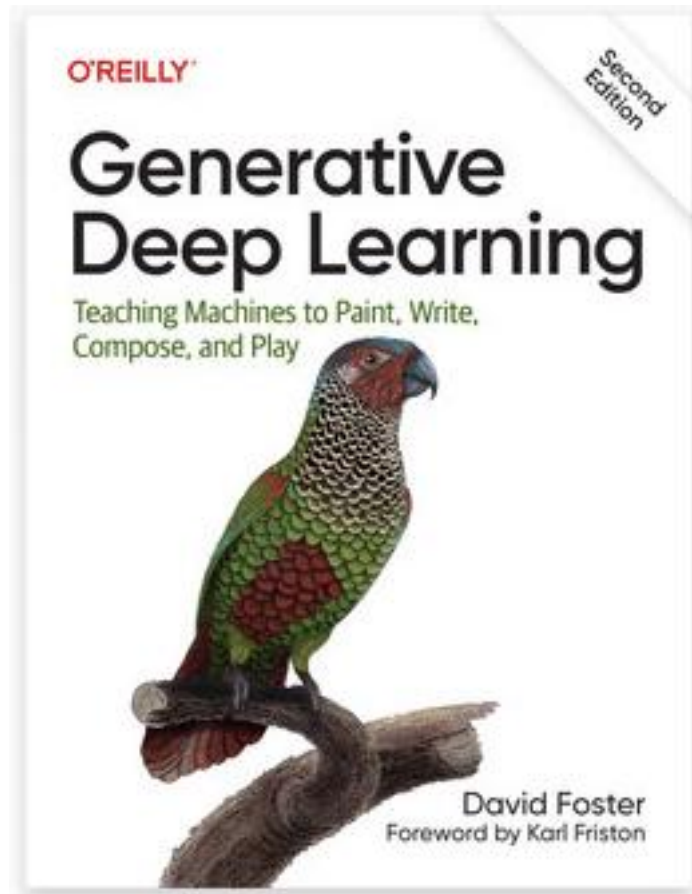
Assessment 2

Assessment 2 – Generative AI business case

- File name must be in the form of: TXX_NAME_UOWID.doc where XX is your tutorial group, NAME is the group leader full name (without space or underscore) and UOWID is your 7-digit UOW ID number (not SIM student ID number).
For example, TO2_BobChew_8080426.doc
- (For the report and presentation slides, write the full name of each group member and their UOW ID at the top of first page)

Don't forget 

This is the book we're using



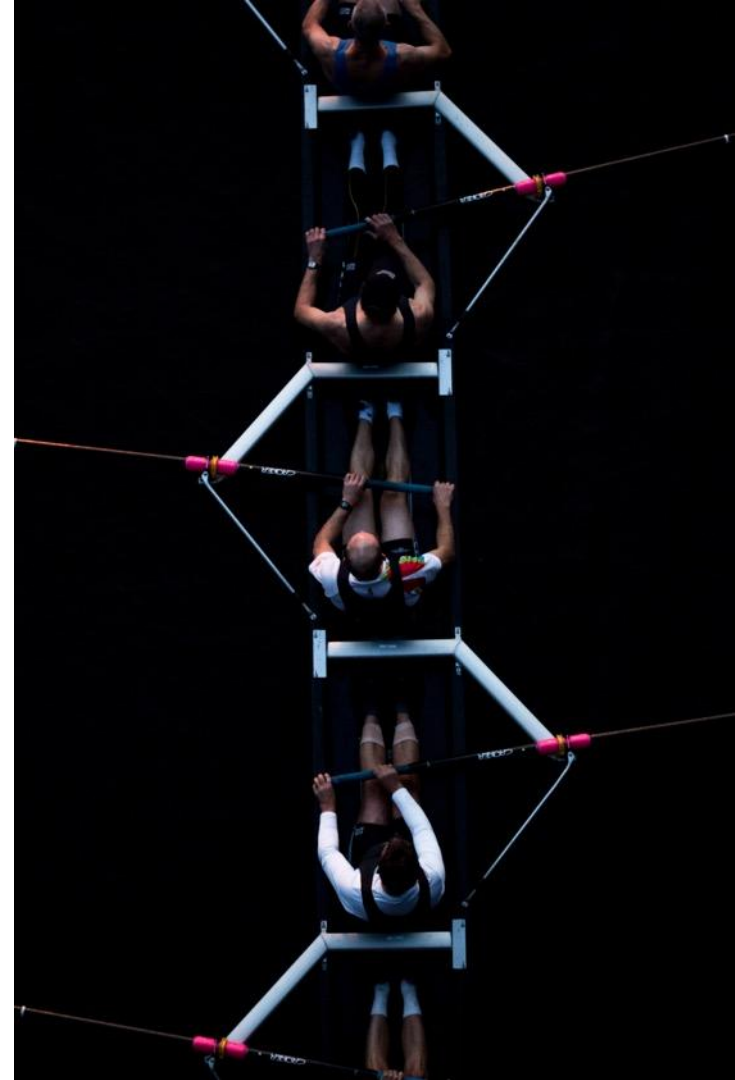
<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

Why are we here?

CSIT205 Generative AI - Module 1

What is Generative AI?

- *Applications of Generative AI*
- *Discriminative models*
- *Generative models*



Disclaimer

The GenAI field is changing very fast

State-of-art in 2024 ≠ state-of-art in 2025

Artificial Analysis

Intelligence Index

Meta Llama 3.3 (Dec '24) 41

Google Gemini 2.0 (Dec '24) 49

OpenAI o3-mini (Dec '24) 66

Deepseek-R1 (Jan '25) 60

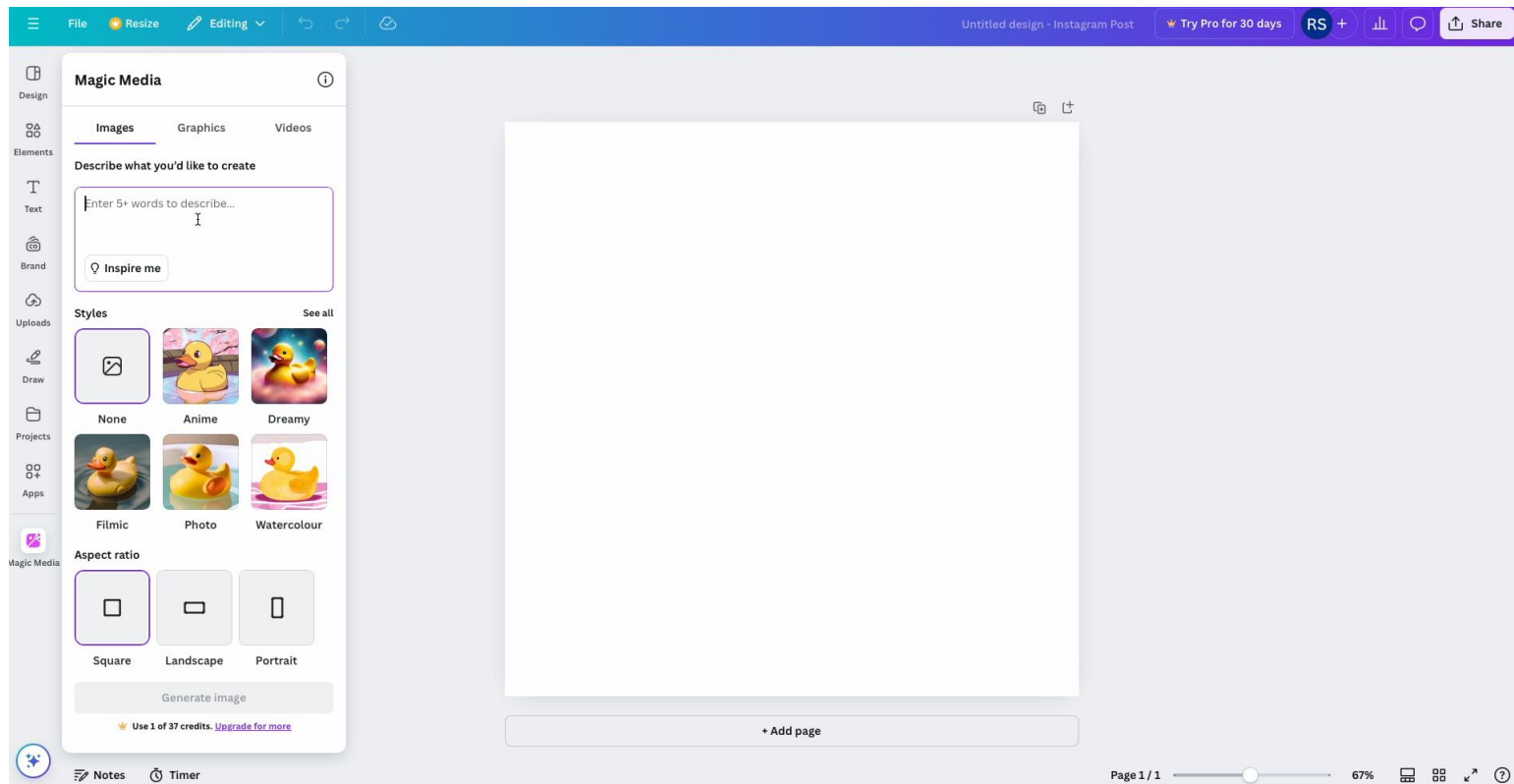
Anthropic Claude 3.7 Sonnet (Feb '25) 57

Practical applications

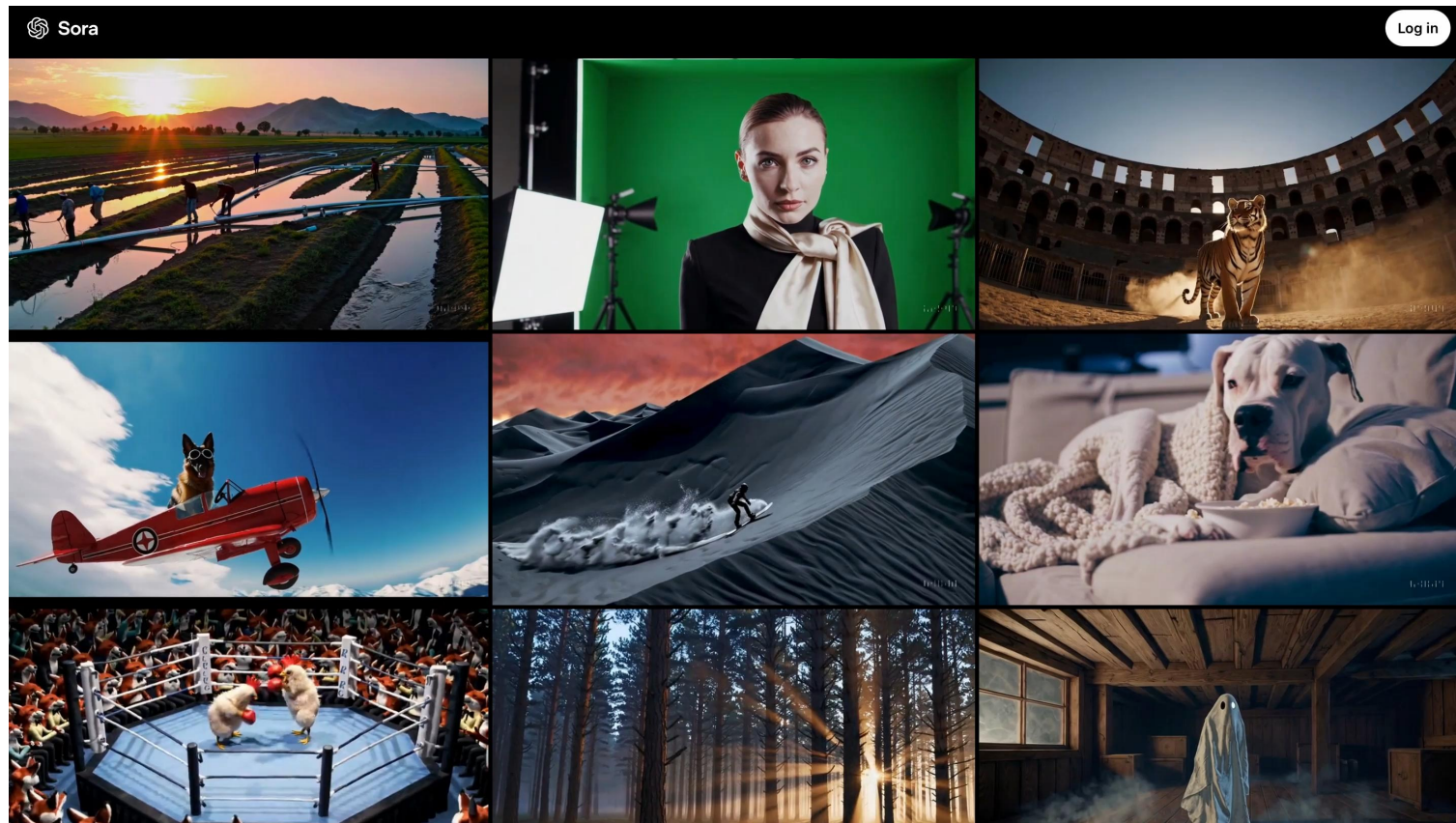


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Generating Faces



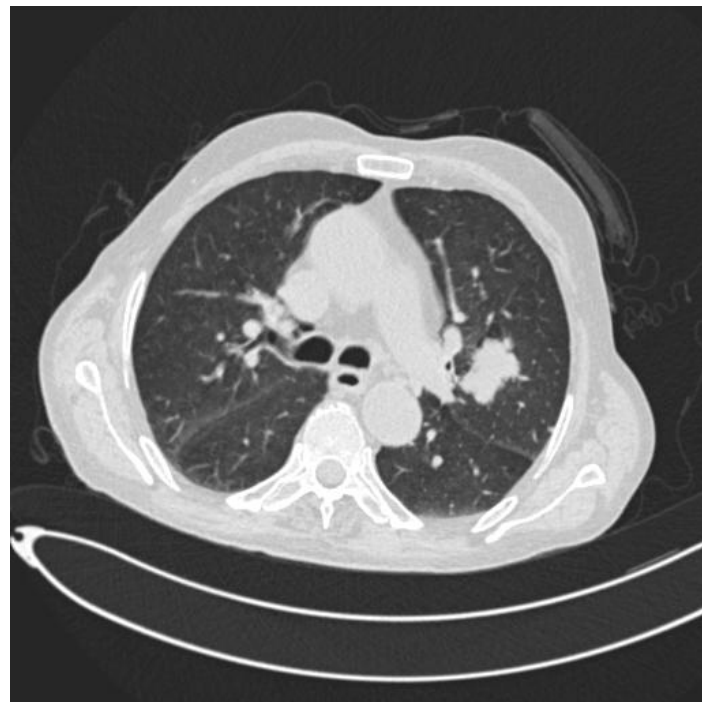
Generating Videos



Lung cancer prediction



Original scan



GenAI

Summarise texts

What is the key topic of this text?

This book gives an introduction to basic neural network architectures and learning rules. Emphasis is placed on the mathematical analysis of these networks, on methods of training them and on their application to practical engineering problems in such areas as nonlinear regression, pattern recognition, signal processing, data mining and control systems.

Every effort has been made to present material in a clear and consistent manner so that it can be read and applied with ease. We have included many solved problems to illustrate each topic of discussion. We have also included a number of case studies in the final chapters to demonstrate practical issues that arise when using neural networks on real world problems.

Since this is a book on the design of neural networks, our choice of topics was guided by two principles. First, we wanted to present the most useful and practical neural network architectures, learning rules and training techniques. Second, we wanted the book to be complete in itself and to flow easily from one chapter to the next. For this reason, various introductory materials and chapters on applied mathematics are included just before they are needed for a particular subject. In summary, we have chosen some topics because of their practical importance in the application of neural networks, and other topics because of their importance in explaining how neural networks operate.

We have omitted many topics that might have been included. We have not, for instance, made this book a catalog or compendium of all known neural network architectures and learning rules, but have instead concentrated on the fundamental concepts. Second, we have not discussed neural network implementation technologies, such as VLSI, optical devices and parallel computers. Finally, we do not present the biological and psychological foundations of neural networks in any depth. These are all important topics, but we hope that we have done the reader a service by focusing on those topics that we consider to be most useful in the design of neural networks and by treating those topics in some depth.

This book has been organized for a one-semester introductory course in neural networks at the senior or first-year graduate level. (It is also suitable for short courses, self-study and reference.) The reader is expected to have some background in linear algebra, probability and differential equations.



The key topic of this text is an introduction to basic neural network architectures and learning rules, with a focus on their mathematical analysis, training methods, and applications to practical engineering problems.

AI-assisted coding

The screenshot displays a development environment with a chat interface on the left and a code editor on the right.

Chat Interface (Left):

- Header: AMAZON Q: CHAT
- Buttons: Chat, X, +, critique this file
- Message: I'll analyze the code and provide a critique focusing on several aspects:
- Critique Points:
 - 1. Security and Configuration**
 - The file path is hardcoded, which reduces flexibility and portability
 - The model parameters (temperature, num_predict) should ideally be in a configuration file
 - The model name "llama3.2" might need verification as it's not a standard name
 - 2. Error Handling**
 - There's no error handling for file loading or LLM operations
 - Missing checks for file existence
 - No handling of potential API errors from the LLM
 - 3. Async Implementation**
 - While async is used, it's not leveraged effectively
 - The `pages.append()` operation inside the async loop could be more efficient
 - The final LLM call is synchronous, not taking advantage of async capabilities
 - 4. Code Structure**
 - Global variables (`llm`, `file_path`, `loader`, `pages`) could be better encapsulated
 - The main functionality could be better organized into a class
- Suggested Improved Version:

```
import os
from pathlib import Path
from typing import List
from langchain_community.document_loaders import PyPDFLoader
from langchain_ollama import ChatOllama
from langchain.schema import Document
```
- Input field: Ask a question or enter "/" for quick actions
- Footer: Amazon Q Developer uses generative AI. You may need to verify responses. See the AWS Responsible AI Policy.

Code Editor (Right):

- File: pdf-ai-summariser
- File: file_loader.py
- Code:

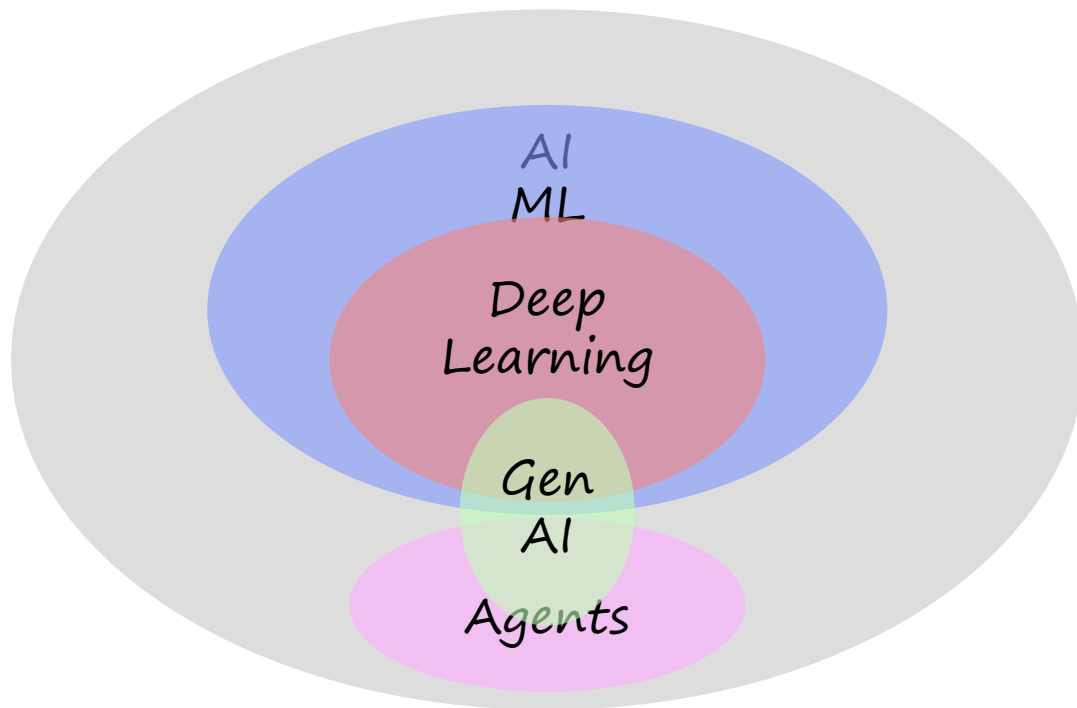
```
1 from langchain_community.document_loaders import PyPDFLoader
2 from langchain_ollama import ChatOllama
3
4 llm = ChatOllama(
5     model="llama3.2",
6     temperature=0.5,
7     num_predict=256,
8 )
9
10 file_path = (
11     "/Users/rorysie/Downloads/NNDesign.pdf"
12 )
13
14 loader = PyPDFLoader(file_path)
15 pages = []
16
17 async def load(file_path: str) -> List:
18     async for page in loader.alazy_load():
19         pages.append(page)
20     return pages
21
22 if __name__ == "__main__":
23     import asyncio
24     pages = asyncio.run(load(file_path))
25     message = llm.invoke("What is the key topic in this text?" + pages[-1].page_content)
26     print(message.content)
```
- Terminal Output:

```
pdf-ai-summariser-py3.12- pdf-ai-summariser poetry run python file_loader.py
The key topic of this text appears to be the implementation or focus on "Good Health and Well-being" as part of the Sustainable Development Goals (SDGs).
pdf-ai-summariser-py3.12- pdf-ai-summariser poetry run python file_loader.py
The key topic in this text appears to be "Competitive Networks" and "Associative Learning", which suggests that the text is a chapter or section within a book or document on artificial neural networks, specifically focusing on types of competitive learning algorithms.
pdf-ai-summariser-py3.12- pdf-ai-summariser poetry run python file_loader.py
The key topic of this text is an introduction to basic neural network architectures and learning rules, with a focus on their mathematical analysis, training methods, and applications to practical engineering problems.
pdf-ai-summariser-py3.12- pdf-ai-summariser
```

Underlying GenAI capabilities

- *Generating Faces: Image generation*
- *Generating Videos: Image generation*
- *Lung cancer detection: Image classification*
- *Lung cancer prediction: Image generation*
- *Summarise texts: Text summarisation*
- *AI-assisted coding (writing code): Text generation*
- *AI-assisted coding (critiquing): Text evaluation*

GenAI vs AI vs Machine Learning vs Deep Learning



Supervised learning

Data Label



Dog



Dog



Dog



Cat



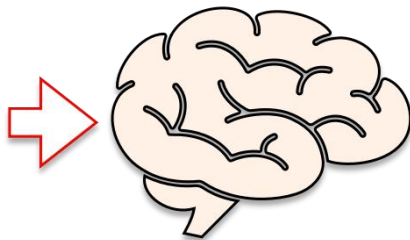
Cat



Cat



'train' a model
to classify



Cat

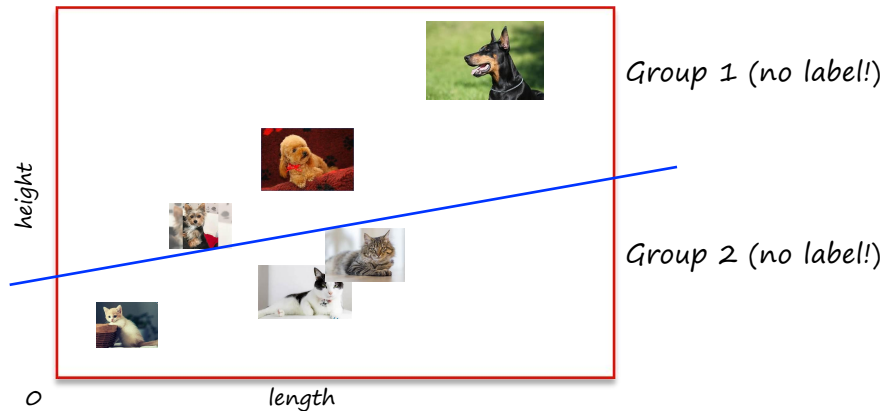
Unsupervised learning

<u>Data</u>	<u>Label</u>
	Dog
	Dog
	Dog
	Cat
	Cat
	Cat

No labels, so we train a model to distinguish groups



New input is put into either group



GenAI is often a form of unsupervised learning

Data



Learn a representation
of these images

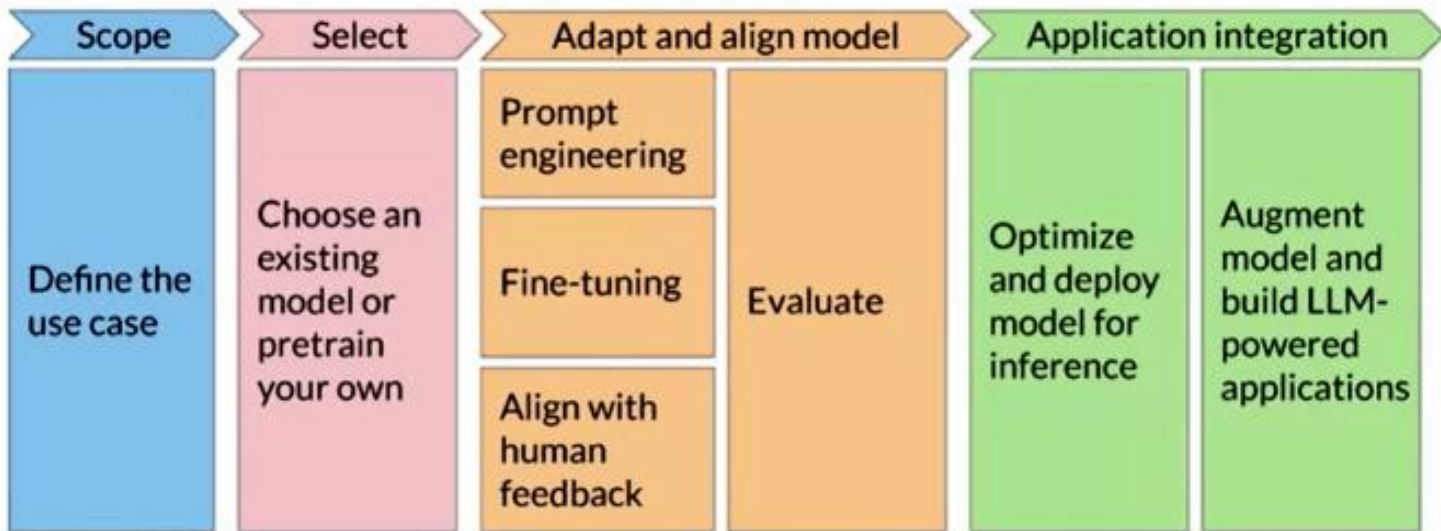
'Create new image'



Why did this fail? We need more data → We need more compute
It's creative 🤖

This is why Generative AI is so challenging

GenAI lifecycle



deeplearning.ai

Module 3: Business Applications and Strategy

Module 1: Generative AI and Assisted Coding

Module 2: Enhancing Model Performance

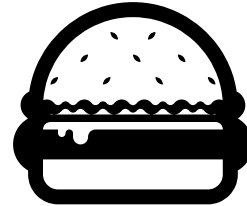
Module 3: Business Applications and Strategy

10-minute Break!

This will help you learn



10 minutes





Which types of Generative AI are these?

- *Generating Faces:*
- *Generating Videos:*
- *Lung cancer detection:*
- *Lung cancer prediction:*
- *Summarise texts:*
- *AI-assisted coding (writing code):*
- *AI-assisted coding (critiquing):*
- *Image generation*
- *Image generation*
- *Image classification*
- *Image generation*
- *Text summarisation*
- *Text generation*
- *Text evaluation*

Discriminative Models

Note: no need to memorise mathematical functions, it is just to build an intuition around GenAI's inner workings



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The Perceptron



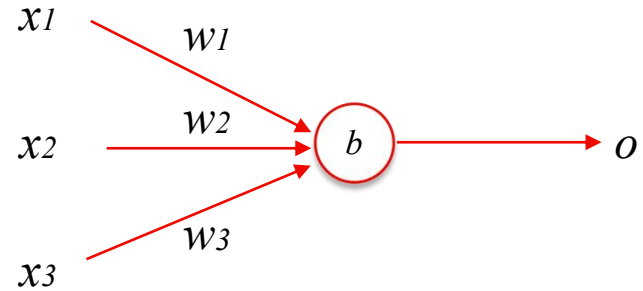
Perceptron

1958

Perceptron

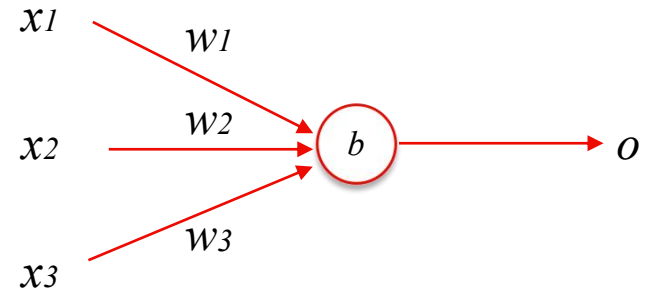
Linear function to model binary outcomes:

- Yes or no ('will I need to bring an umbrella?')
- True or False
- 0 or 1



Perceptron - how it works

- Linear function
 - Several inputs x
 - Weights w
 - Bias b
 - One output o
- Learns by updating its weights w and bias b
- Supervised learning



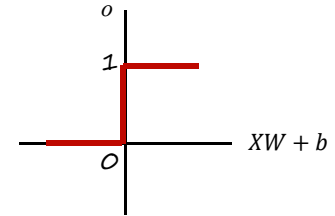
Perceptron (video)

- https://youtu.be/OFbnpY_k7js?si=B_nCso1-boByVoJB

The Activation function is a Step function

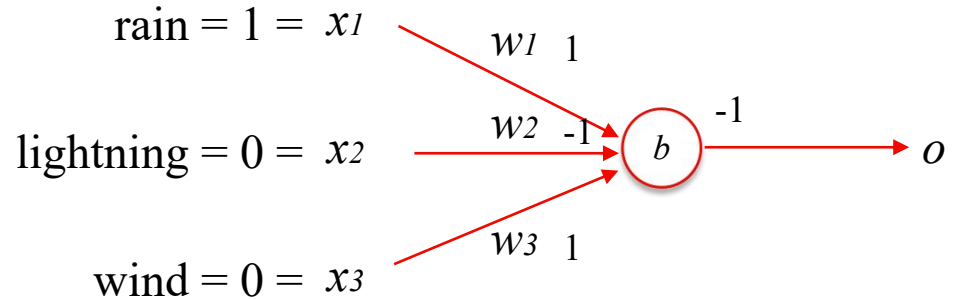
- Binary classification $[0,1]$

$$o = \begin{cases} 0 & \text{if } \sum_{j=1}^n x_j w_j + b \leq 0 \\ 1 & \text{if } \sum_{j=1}^n x_j w_j + b > 0 \end{cases}$$



Perceptron - example

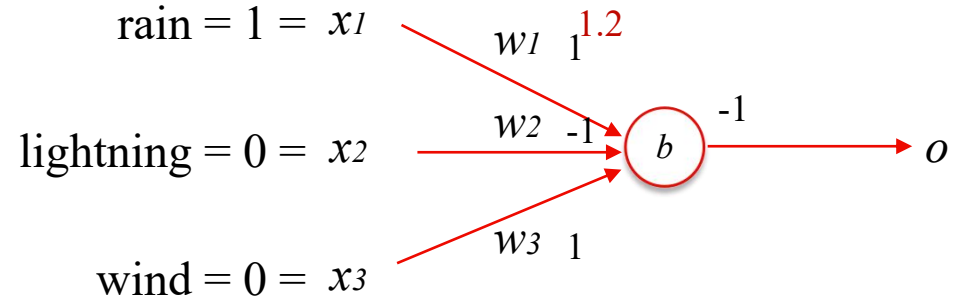
- 0 is false; 1 is true
- We know that in this case the output o should be 1.
- Assign random weights W and bias b and evaluate



result = 0, so update the weights

Perceptron - learning rule

- $w_j \leftarrow w_j + \Delta w_j$
- $\Delta w_j = \eta(t - o)x_j$
- η is the learning rate, say 0.2
- t is the target 1
- o is the actual output o
- $\Delta w_1 \leftarrow 0.2 * (1 - 0) * 1 = 0.2$
- $w_1 = w_1 + 0.2 = 1.2$

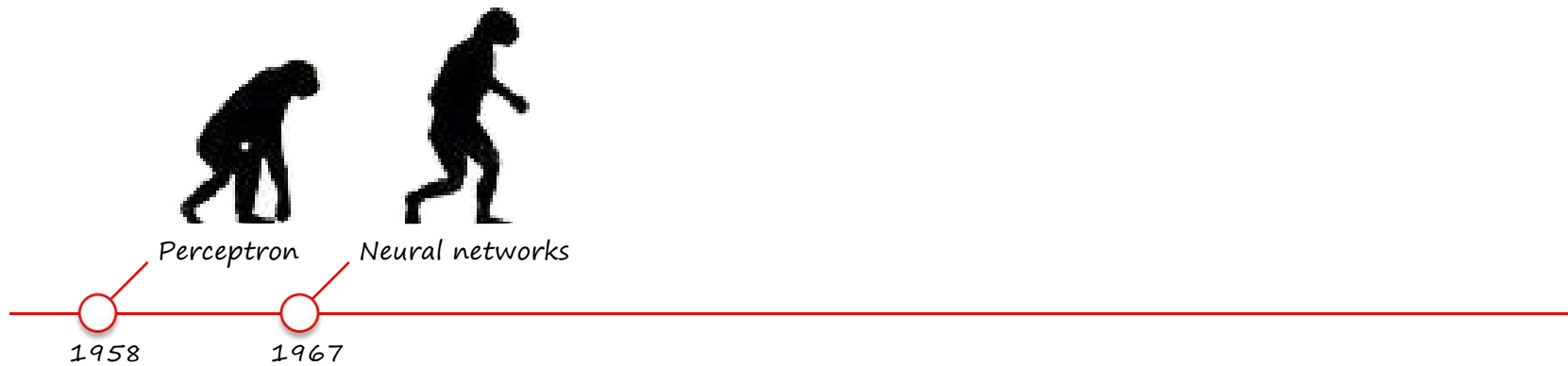


Limitations of perceptrons

- ✗ *Only binary classification $[0,1]$*
- ✗ *Single layer is too simple to represent complex problems; limited learning capabilities*

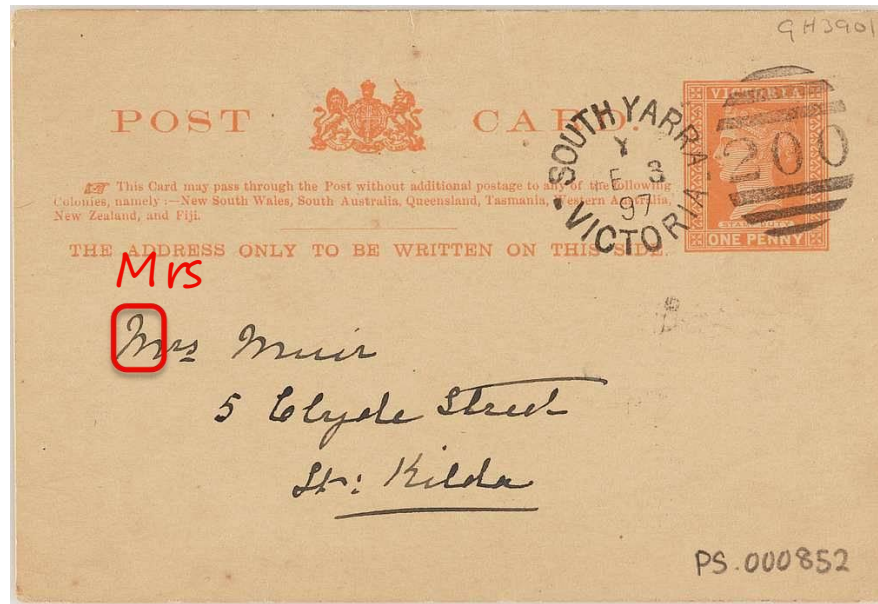


Neural Networks



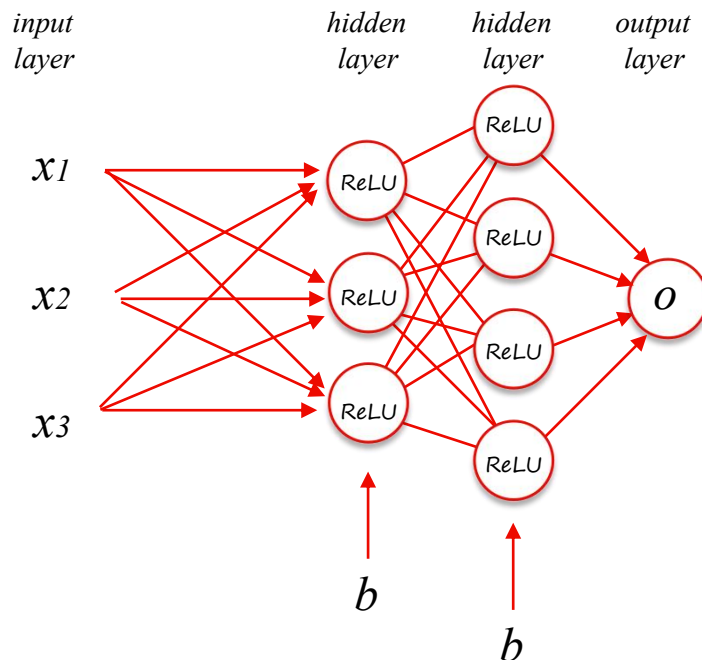
Neural networks

- Multi-Layer Perceptron that learned weights by Amari (1967)
- Used stochastic gradient descent
- Applied to for instance object character recognition (OCR)



Neural networks

- Basically a perceptron with more hidden layers:
 - One input layer
 - One output layer
 - Multiple hidden layers
- Weights (edges) and biases between layers are updated while learning
- Calculations go from left to right ('feed-forward neural network')



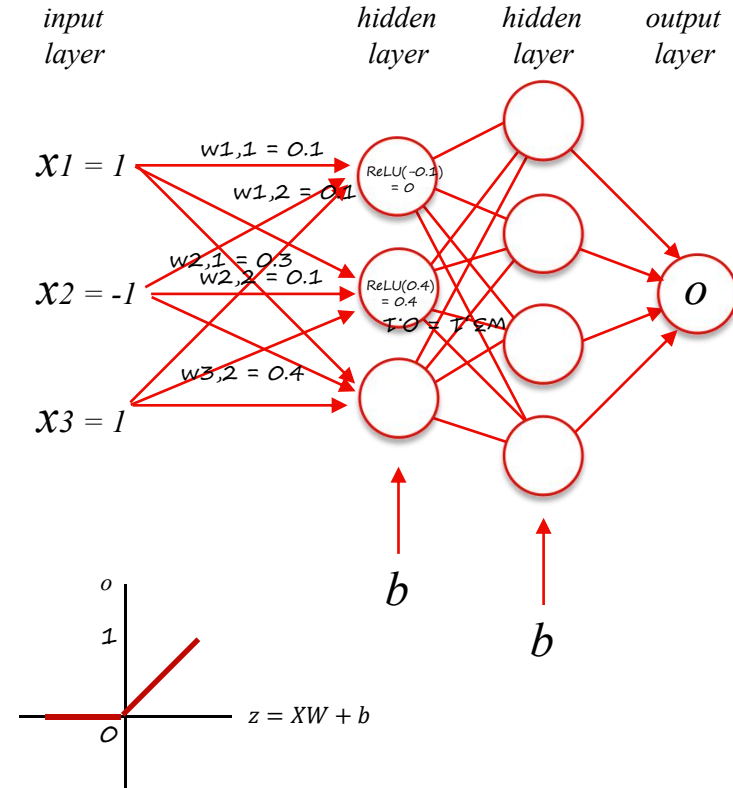
Activation function is not a step function, but ReLU

Rectified Linear Unit (ReLU)

- The step function only outputs 0 or 1, which is problematic in larger networks.
- So we use a Rectified Linear Unit (ReLU) that allows for more gradual changes in the network.

$$o = \max(0, z)$$

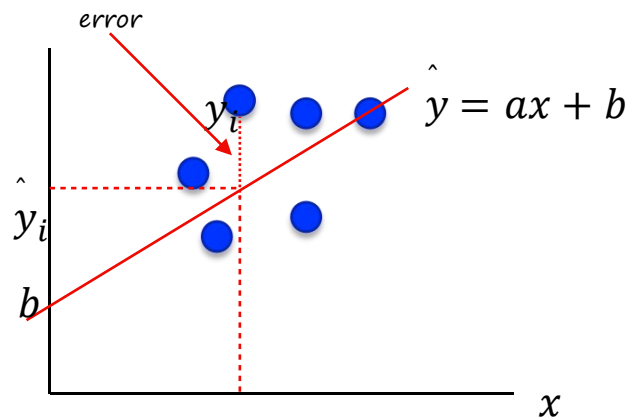
$$o = \begin{cases} 0 & \text{if } \sum_{j=1}^n x_j w_j + b \leq 0 \\ z & \text{if } \sum_{j=1}^n x_j w_j + b > 0 \end{cases}$$



Neural Networks: Learning rule

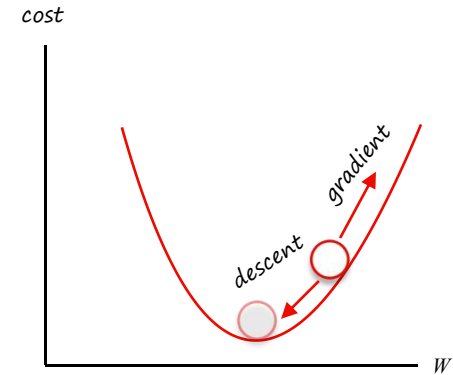
- Loss function $C(w)$ (e.g. RMSE) calculates the error / cost for predictions $\hat{y}$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



Neural Networks: Learning rule

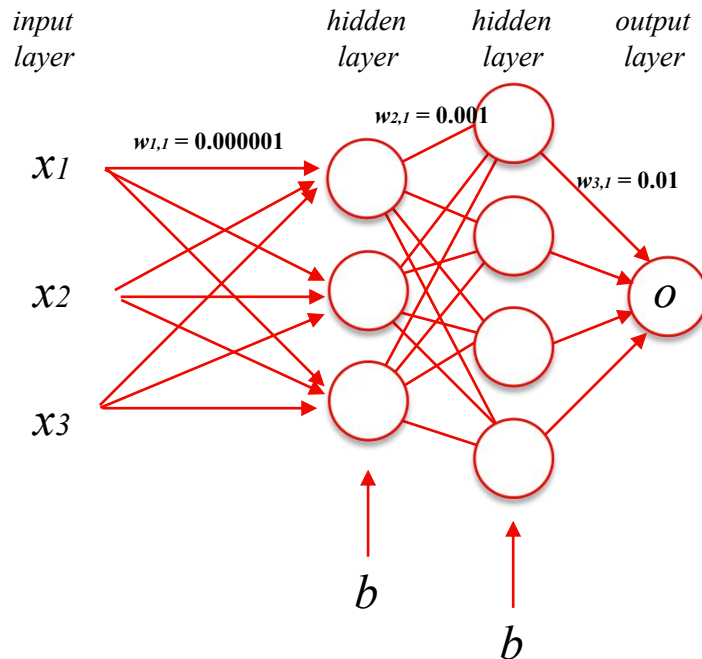
- Back-propagation ('backprop') is used to determine the 'gradient', i.e. the speed at which the cost increases as you adapt weights and go backward in the network.
- Stochastic Gradient Descent learns by minimising the cost, i.e. descending along the gradient, and it does so by taking a sample ('stochastic').



Limitations of Neural Networks

✗ *Vanishing / exploding gradient:*
weights can become increasingly smaller
/ greater as you move backward in the
network (e.g. 0.01^n).

✗ *Cannot handle sequence data (e.g.
“The quick brown fox jumps over the
lazy dog”)*

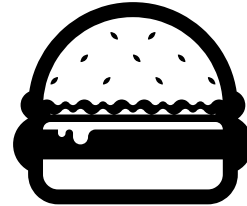


10-minute Break!

This will help you learn



10 minutes





What is the ReLU function?

A. $o = \max(0, z)$

B. $o = 0 \text{ or } 1$

C. $o = \max(0, 1)$

D. $o = \min(0, z)$

What is not a limitation of Neural Networks?



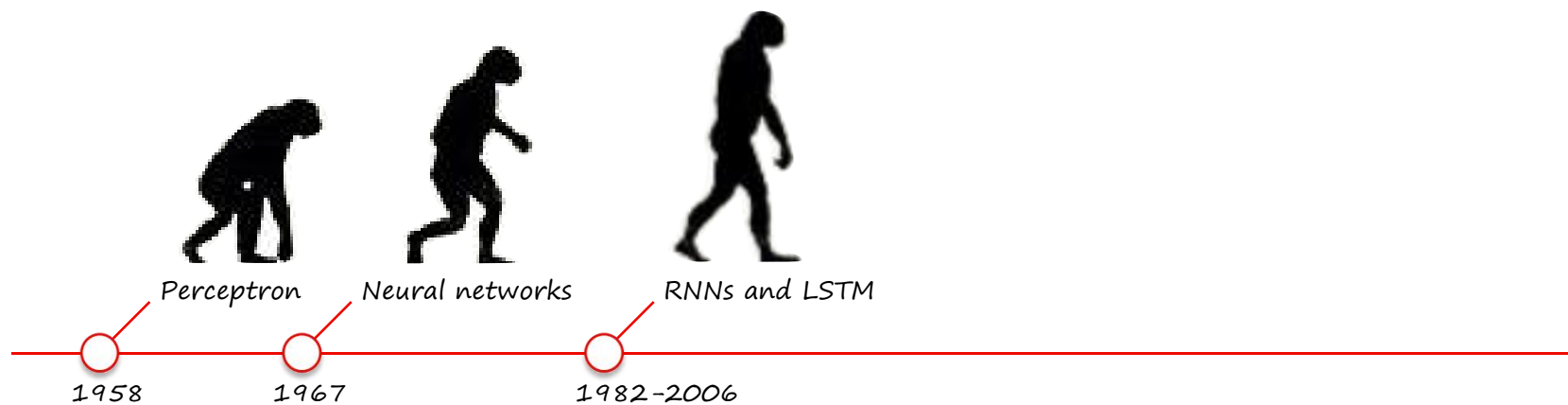
- A. Vanishing gradients
- B. Exploding gradients
- C. Inability to handle sequence data
- D. Can only handle binary classification $[0,1]$

Generative Models



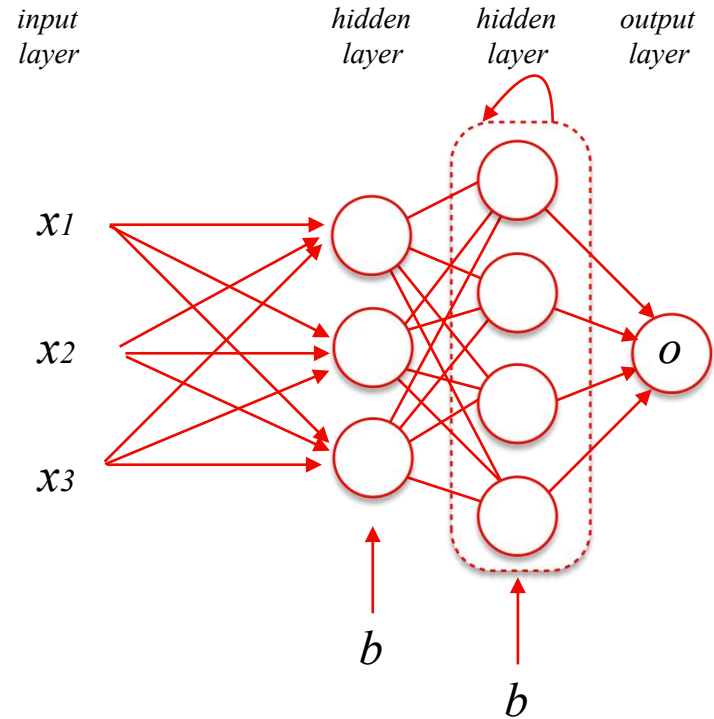
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RNNs and LSTM



Recurrent Neural networks

- Basically NNs that can have loops to / within their own layer
- So they can model relationships forward and backward in time (e.g. sequences): “The quick brown fox jumps over the lazy dog”

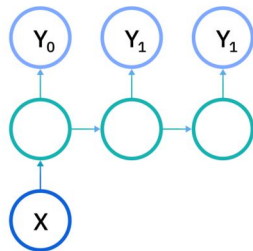


Many types of RNNs

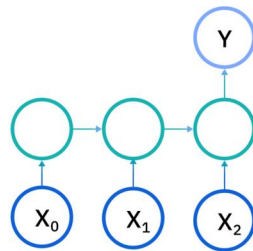
One-to-one



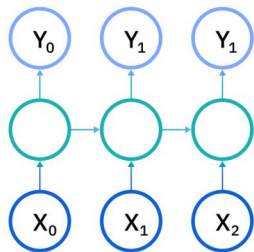
One-to-many



Many-to-one

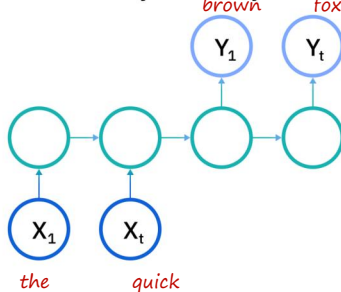


Many-to-many



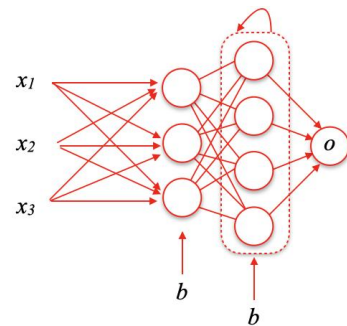
Translation

Many-to-many



Text generation

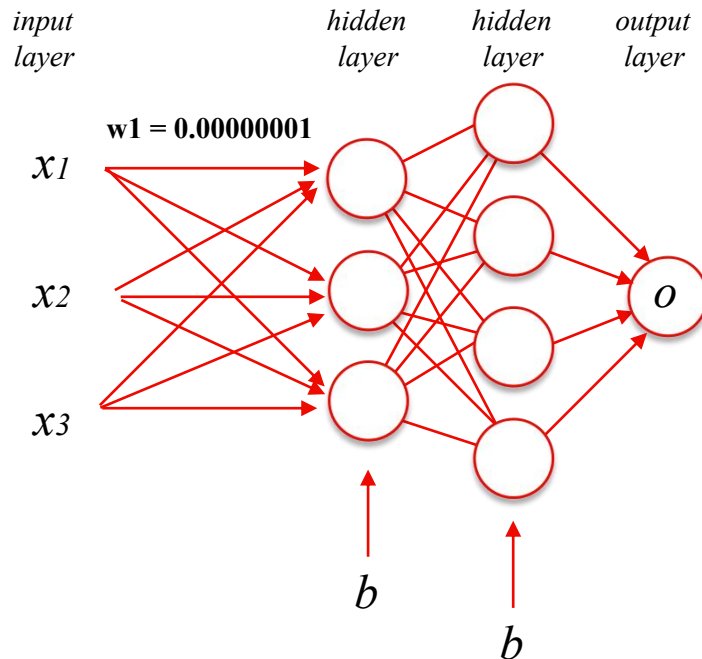
'unfold'



<https://machinelearningmastery.com/rnn-unrolling/>

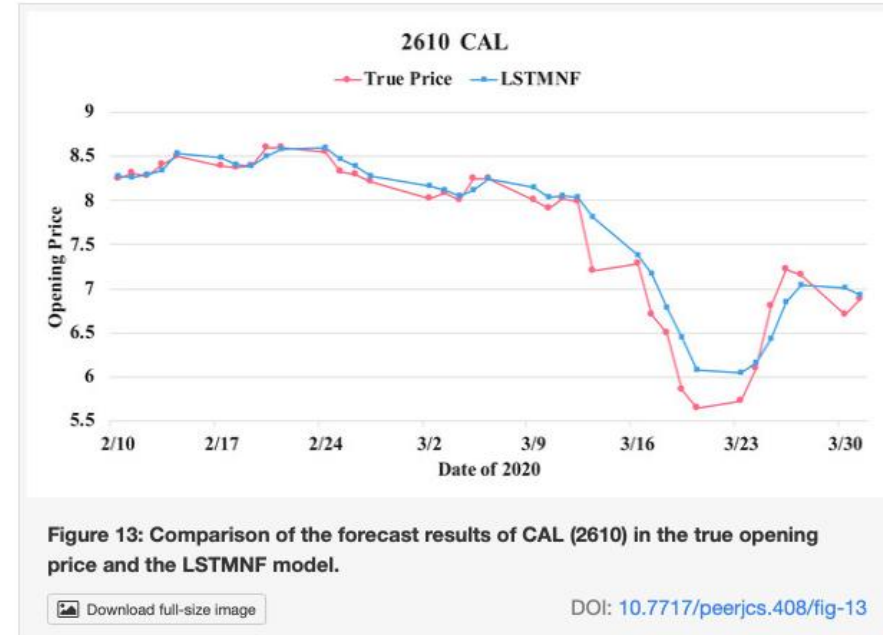
RNNs: Limitations

- ✗ *Vanishing gradient problem*
- ✗ *Long term dependency problem:*
longer sequences are challenging, because
all information is kept.



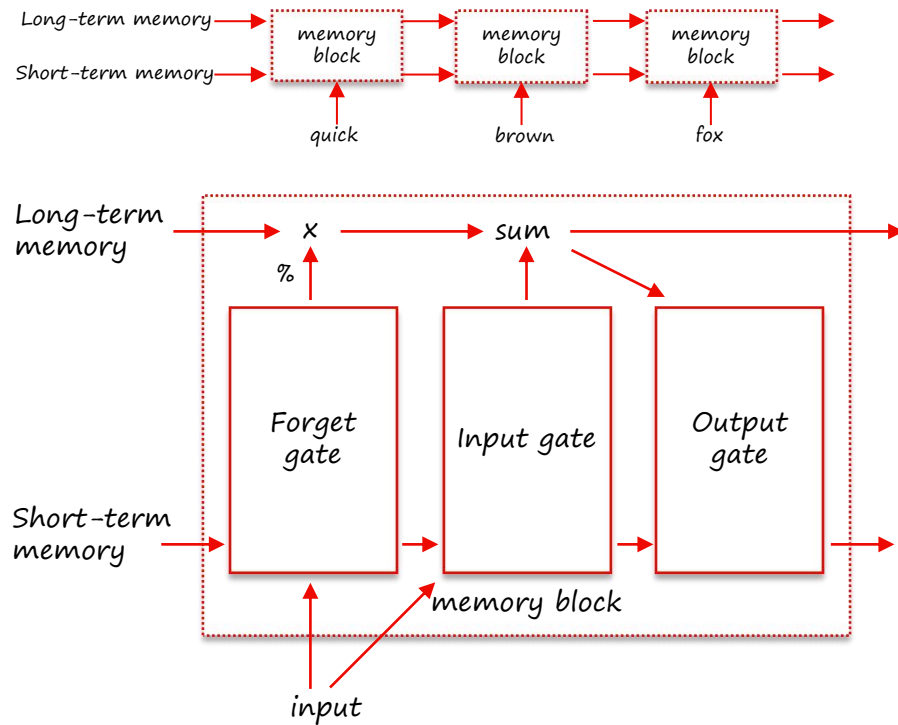
Long Short-Term Memory (LSTM)

- Type of RNN
- Used for:
 - Language understanding:
Google BERT (Devlin et al., 2019)
 - Time series forecasting (Siarni-Namini et al., 2018)
 - etc.



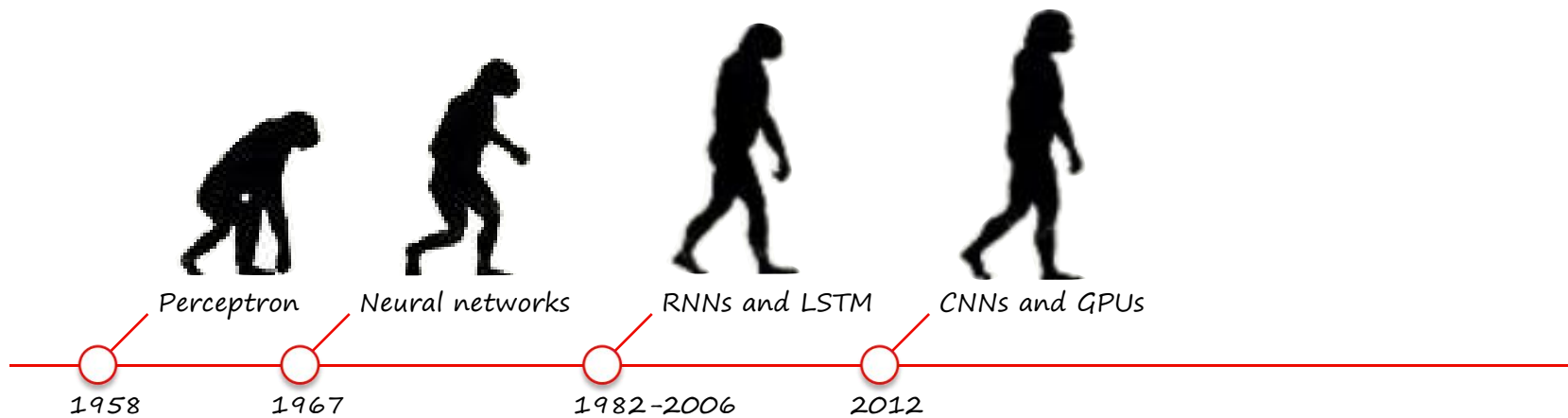
Long Short-Term Memory (LSTM)

- *Memory blocks instead of neurons*
- *Employs a forget gate that transforms input between 0 and 1, which means it can retain important information*
- *→ addresses the vanishing gradient and long term dependency problems (but not fully)*



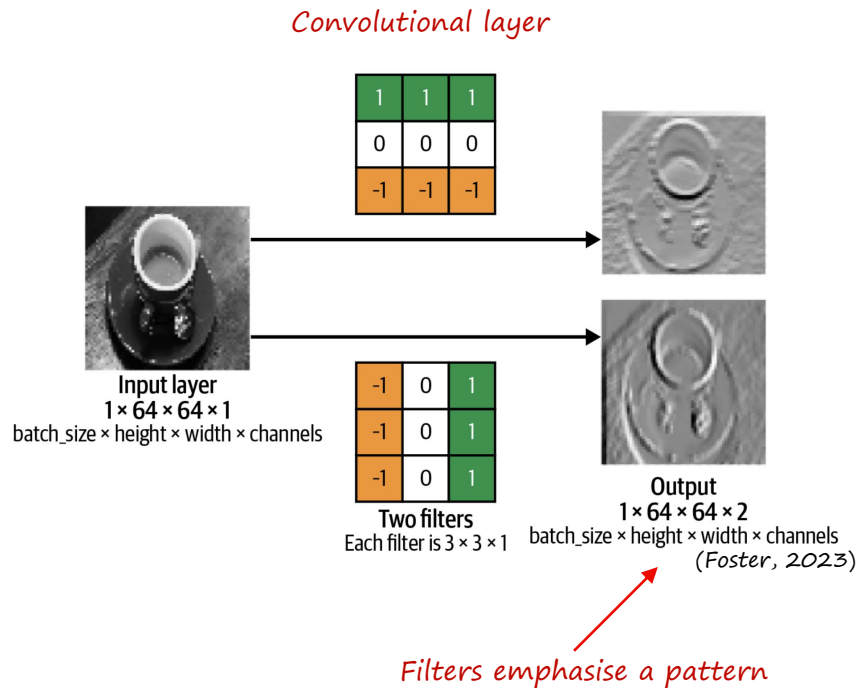
More info: <https://youtu.be/YCzL96nL7j>

Deep Learning: CNNs and GPUs



Convolutional Neural Networks (CNNs)

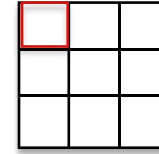
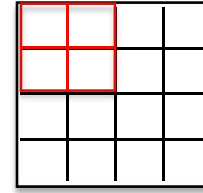
- 2012: AlexNet (Krizhevsky et al., 2012) wins ImageNet competition.
- Analyses grid-like topologies, like images.
- Architecture:
 - Convolutional layers that apply kernels / filters to detect features
 - Pooling layers (not shown) to reduce dimensionality



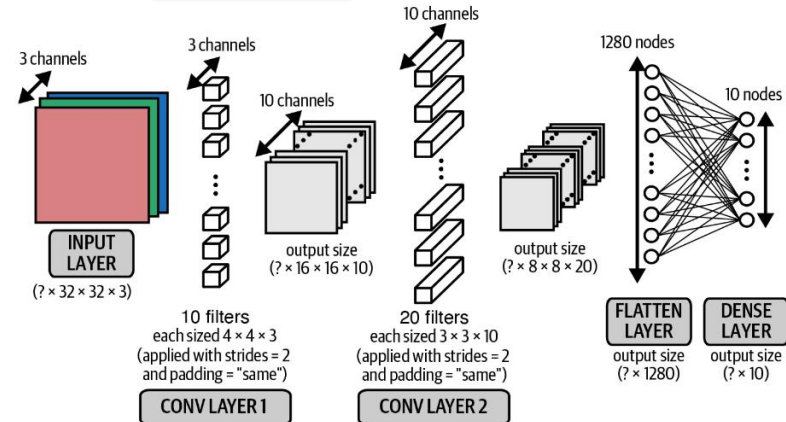
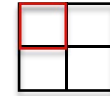
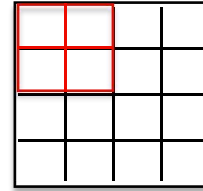
CNNs in more detail

- Strides make the filters move along the input. Stride = 2 essentially halves the input shape.
- 3 colour channels $\rightarrow$ filters take shape height $\times$ width $\times$ 3
- Output dimension of $h \times w \times 10$ (see image on the right) = $h \times w \times$ number of filters.

2×2 filter with stride = 1



2×2 filter with stride = 2



CNNs in more detail

- Pooling layers reduce dimensionality

2 x 2 max pooling with stride 2

0.8	0.9	0.1	0.1
0.2	0.4	0.5	0.8
0.2	0.5	0.7	0.1
0.2	0.5	0.4	0.3

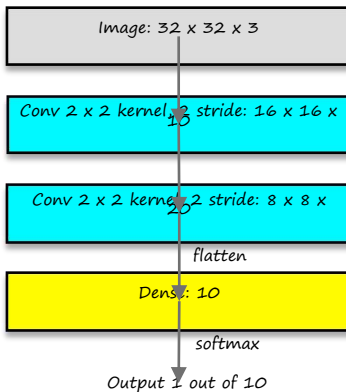
0.9	0.8
0.5	0.7

Usually presented like this

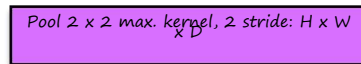
Layer (type)	Output shape	Param #
InputLayer	(None, 32, 32, 3)	0
Conv2D	(None, 16, 16, 10)	490
Conv2D	(None, 8, 8, 20)	1,820
Flatten	(None, 1280)	0
Dense	(None, 10)	12,810

Total params 15,120
 Trainable params 15,120
 Non-trainable params 0

Or this



FYI pooling is represented like this:



CNNs can be trained on GPUs

- Graphics Processing Units were originally used for computer graphics.
- 2006: NVIDIA releases CUDA, which opened up GPUs for general purpose processing.
- 2006: Chellapilla, et al. train a CNN on GPU that is 3.11 - 4.11 times faster than on CPU.
- 2012: Cireşan et al. train a CNN on GPU that is 60 times faster than on CPU.



<https://www.nvidia.com/en-us/data-center/h100/>

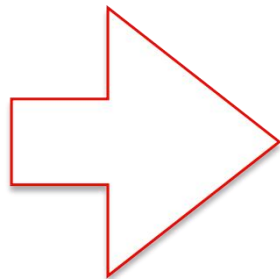
Wrap-up



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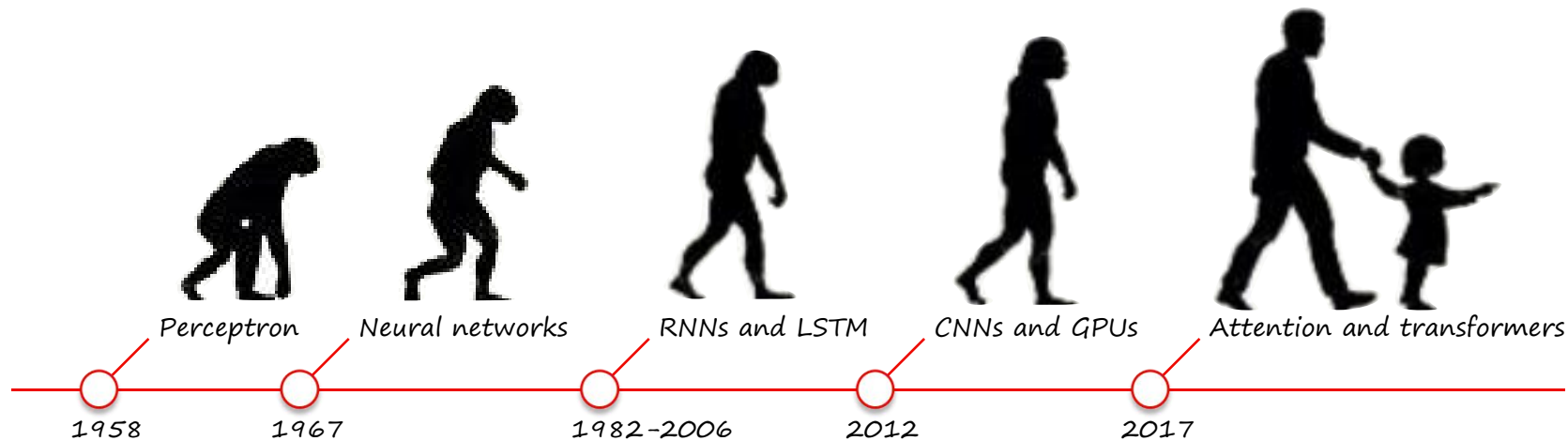
Summary

- We have practical applications that translate text-to-text, text-to-image, image-to-image, etc.
- Early AI models were mostly discriminative: they assigned a class / score to the input data
- Later AI models like RNNs started becoming generative: they created output based on certain input text (e.g. translation)
- GPUs significantly accelerated model training, and thus boosted AI research.



Deep Neural
Networks training
was scalable

Next lecture: Attention and transformers



Homework

- *Read Foster (2023): Chapter 3 and 4*

Take Home Message...

*Consider the message that you want to create when
designing your interfaces.*



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Things to think about

What is the purpose of prototyping?

*What is the difference between Lo-fi
and Hi-fi designs?*

*What is a storyboard? When can
these be used?*

What should you show your client?

References and Resources

- Amari, S. (1967). A Theory of Adaptive Pattern Classifiers. *IEEE Transactions on Electronic Computers*, EC-16.
- Chellapilla, K., Puri, S., & Simard, P. (2006). High Performance Convolutional Neural Networks for Document Processing. Tenth International Workshop on Frontiers in Handwriting Recognition. <https://inria.hal.science/inria-00112631v1>
- Cireşan, D. C., Meier, U., Masci, J., Gambardella, L. M., & Schmidhuber, J. (2011). High-Performance Neural Networks for Visual Object Classification. <http://arxiv.org/abs/1102.0183>
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6), 84–90. <https://doi.org/10.1145/3065386>
- Rosenblatt, F. (1958). THE PERCEPTRON: A PROBABILISTIC MODEL FOR INFORMATION STORAGE AND ORGANIZATION IN THE BRAIN. *Psychological Review*, 65(6), 19–27.
- Siami-Namini, S., Tavakoli, N., & Siami Namin, A. (2018). A Comparison of ARIMA and LSTM in Forecasting Time Series. *Proceedings - 17th IEEE International Conference on Machine Learning and Applications, ICMLA 2018*, 1394–1401. <https://doi.org/10.1109/ICMLA.2018.00227>

Questions



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